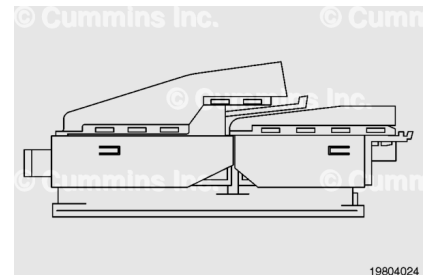


Trainings ▾

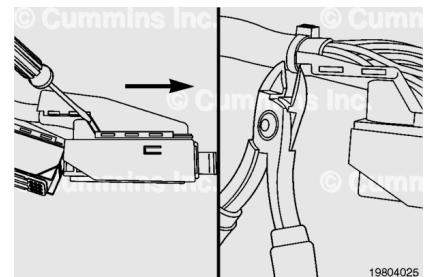
Pin Replacement

This connector is used to attach the appropriate harness to the DEF (Diesel Exhaust Fluid) Controller.



Remove the connector cover by inserting a small screwdriver into the angled slot on the side of the cover and twisting. Then slide the cover off of the connector.

Cut the wire tie holding the wire bundle to free the wires.

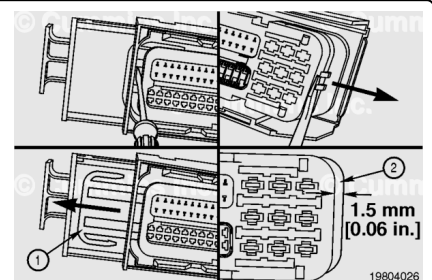


53 Pin Connector

Before the 1.2mm terminals can be removed it is necessary to remove the secondary lock #1 from the small terminal block using a small screwdriver to move from the closed detent and completely remove from connector body.

Before the 2.8mm terminals can be removed it is necessary to unlock the secondary lock #2 by using a small screwdriver to move secondary lock #2 from closed detent to the open detent position approximately 1.5mm travel.

DO NOT REMOVE COMPLETELY.

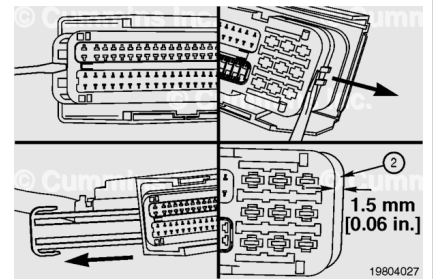


86 Pin Connector

Repeat steps from 53 way connector.

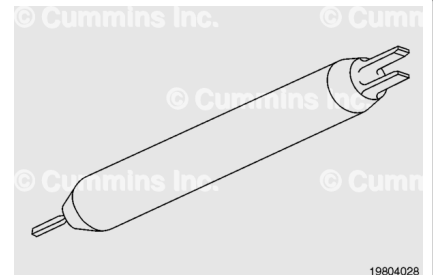
Note : The secondary locks are on opposite ends of the 86 way connector than the 53 way connector

Notice that the secondary lock #2 should only move about 1.5mm and stop in an open detent position.



The same procedure is used to remove the wires from both connectors.

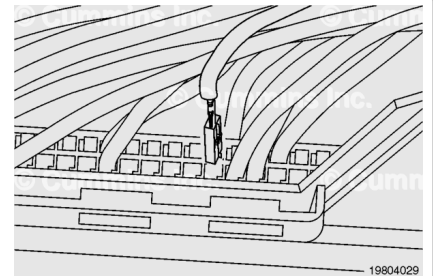
Use Cummins electrical terminal replacer part number 4919735 to remove the wire terminals from the connector.



Replace one terminal wire at a time. If more than one terminal wire must be replaced, attach an identification tag to each wire removed.

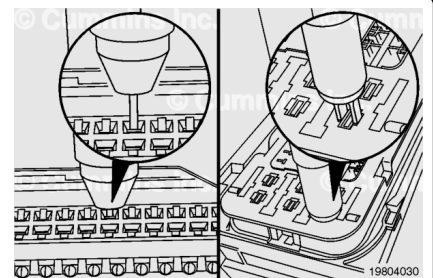
Refer to the wiring diagram in Section E for terminal locations.

Refer to the appropriate wiring harness repair kit in the service tools table in front of section 19 for the correct repair wire.



Removing Terminal

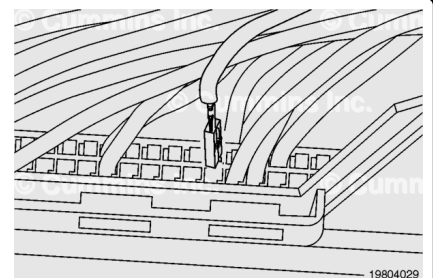
Insert the proper end of the replacer into the terminal unlocking holes in the face of the connector being careful to hold the replacer perpendicular to the face of the connector.



⚠ CAUTION ⚠

If the wire is difficult to remove, do not pull hard on the wire, otherwise, the locking tang of the terminal will stick or the terminal will pull off the wire and remain in the connector.

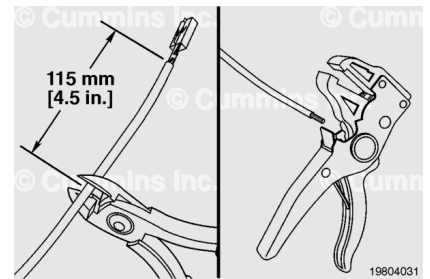
Carefully pull the wire from the connector. If it is difficult to remove, repeat the entire process.



Note : The repair wire is 115mm [4.5 in] long.

Use wire cutters to remove 115mm [4.5 in.] of the terminal and wire to be replaced.

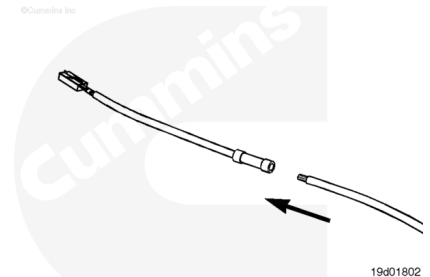
Use wire stripping tool, Part Number 3400045, to remove 6mm [¼ in.] of insulation from the wire.



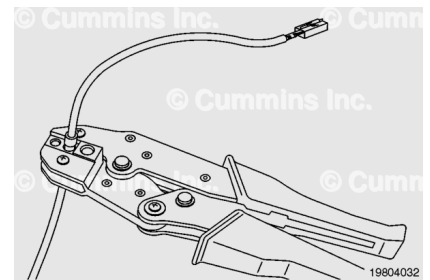
Before installing the new repair wire, perform a test fit to make sure the wire is the correct size.

Install the repair wire on the bare wire.

Make sure the bare wire extends into the splice connector properly.

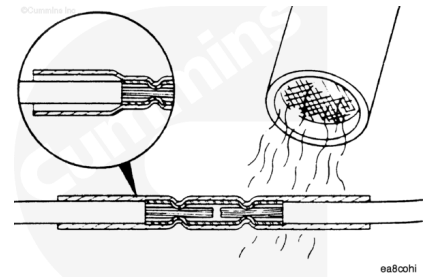


Use wire crimping tool, Part Number 3163109, to crimp the repair wire onto the bare wire.



Use Heat Gun, Part Number 3822860, to heat the shrink tubing around the wire.

The tubing will shrink and make the connection waterproof.



Inserting Terminal

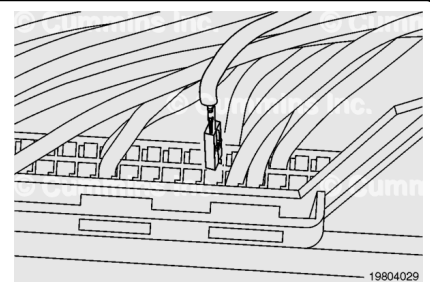
The wire terminals have locating features that only allow the terminal to be inserted in the proper orientation.

Insert the wire from the top of the connector.

Push the wire until the terminal locks into place.

Pull on the wire gently to make sure it is locked into the connector.

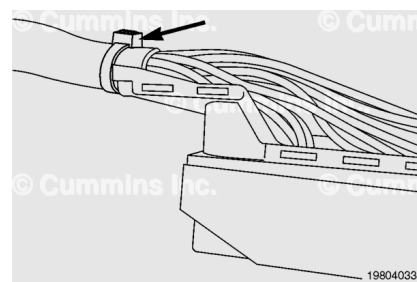
Reinstall secondary locks and snap into locked position.



Position the wire bundle into place.

Install a wire tie to hold the wire bundle in place on the connector.

Remove the excess wire tie.



Replace the connector shell by inserting the bottom sliding guides into the connector guide rails.

Slide toward the wire bundle and align the top slide guides into the upper connector guides.

Slide toward the wire bundle until the connector cover is snapped into place.

